

Random Signal Theory–EEE 554

Lecture Notes 2

Discrete and Continuous Random Variables

[Reading: Kay, Sections 5.3, 5.4, 5.8, 10.3, 10.4, 10.6]

Probability Density Function (PDF) for continuous RV (CRV) [Kay, Section 10.4]

Interpretation:

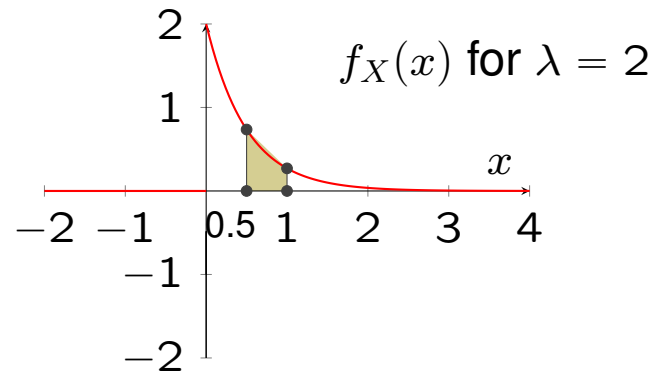
PDF provides the likelihood of a CRV falling within a range of values

⇒ obtain probability values by **integrating PDF**

For CRV X , PDF is $f_X(x) = \frac{d}{dx}F_X(x)$

Example: message transmission time CRV X (see page 8)

since CDF is $F_X(x) = (1 - e^{-2x}) u(x)$, PDF is $f_X(x) = \frac{d}{dx} F_X(x) = 2 e^{-2x} u(x)$



Use PDF to find probabilities: $\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1) = \int_{x_1}^{x_2} f_X(x) dx$

For example:

$$\Pr(0.5 < X < 1) = \int_{0.5}^1 2 e^{-2x} dx = -e^{-2x} \Big|_{0.5}^1 = (e^{-1} - e^{-2}) = 0.2325 \text{ (shaded area)}$$

or equivalently

$$\Pr(0.5 < X < 1) = F_X(1) - F_X(0.5) = (1 - e^{-2}) - (1 - e^{-1}) = (e^{-1} - e^{-2}) = 0.2325$$

Properties of PMF (for discrete RV) and PDF (for continuous RV)

P1. For PMF and PDF: $f_X(x) \geq 0$ non-negative function

Expected since it is the derivative of a monotonically increasing function

P2. For PMF and PDF: $F_X(x) = \int_{-\infty}^x f_X(\gamma) d\gamma$

CDF in terms of PDF (or PMF) (area under the plot)

P3. For PMF and PDF: $\int_{-\infty}^{\infty} f_X(x) dx = 1$

(area under all sample space should be 1)

P4. Obtain probability of values within region

DRV X with PFM $f_X(x)$: $\Pr(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} f_X(x) dx$

CRV X with PDF $f_X(x)$: $\Pr(x_1 < X \leq x_2) = \Pr(x_1 < X < x_2) = \int_{x_1}^{x_2} f_X(x) dx$

Probability density function for mixed RVs

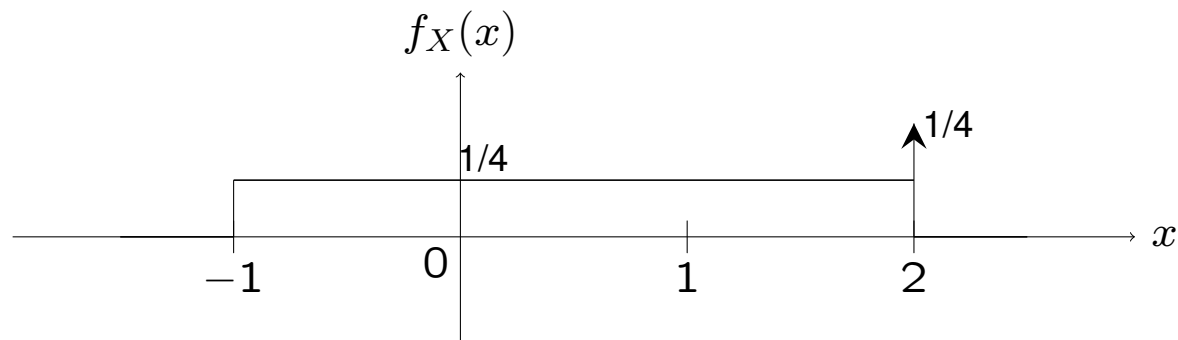
CDF discontinuity at $x = x_i \in \mathbb{Z}$ becomes PDF delta function at $x = x_i$ (derivative relation)

Example (from page 12): Given CDF

$$F_X(x) = \left(\frac{x+1}{4}\right) \left(u(x+1) - u(x-2)\right) + u(x-2) = \begin{cases} 0, & x < -1 \\ (x+1)/4, & -1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

CDF has discontinuity at $x = 2$ of size $\Pr(X = 2) = \frac{1}{4}$ and since $\frac{d}{dx}u(x) = \delta(x)$

$$\begin{aligned} f_X(x) &= \frac{d}{dx}F_X(x) = \frac{1}{4} \left(u(x+1) - u(x-2)\right) + \left(\frac{x+1}{4}\right) \left(\delta(x+1) - \delta(x-2)\right) + \delta(x-2) \\ &= \frac{1}{4} \left(u(x+1) - u(x-2)\right) + 0 - \frac{3}{4}\delta(x-2) + \delta(x-2) \\ &= \frac{1}{4} \left(u(x+1) - u(x-2)\right) + \frac{1}{4}\delta(x-2) = \begin{cases} \frac{1}{4}, & -1 \leq x < 2 \\ \frac{1}{4}\delta(x-2), & x = 2 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$



Note that in previous calculation, also used the Dirac delta function property

$$g(x) \delta(x - x_0) = g(x_0) \delta(x - x_0)$$

$$\Rightarrow \left(\frac{x+1}{4}\right) \delta(x + 1) = \left(\frac{-1+1}{4}\right) \delta(x + 1) = 0$$

How do we check if this is a valid PDF?

$$f_X(x) = \frac{1}{4} \left(u(x + 1) - u(x - 2) \right) + \frac{1}{4} \delta(x - 2)$$

Integration over all values in sample space should equal 1, $\int_{S_X} f_X(x) dx = 1$

In this case,

$$\begin{aligned} & \int_{-\infty}^{\infty} \left(\frac{1}{4} \left(u(x + 1) - u(x - 2) \right) + \frac{1}{4} \delta(x - 2) \right) dx \\ &= \int_{-1}^2 \frac{1}{4} dx + \frac{1}{4} \int_{-\infty}^{\infty} \delta(x - 2) dx = \frac{x}{4} \Big|_{-1}^2 + \frac{1}{4} = 1 \end{aligned}$$